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Caementiciae Structurae: Ancient Roman Concrete Structures Fabricated with Reactive Volcanic Rock

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Annu. Rev. Earth Planet. Sci. 2026. 54:269–99

The *Annual Review of Earth and Planetary Sciences* is online at earth.annualreviews.org

https://doi.org/10.1146/annurev-earth-032524-014538

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Keywords

Roman architectural and marine concrete, volcanic pozzolans, hydrated silicate binding phases, toughness, self-sustaining concrete systems

Abstract

Roman architectural and marine concrete structures, *caementiciae structurae*, have remained intact and functional for two millennia. A mortar fabricated with scoriaceous or pumiceous volcanic tephra aggregate (*harenae fossiciae*, *pulvis*) binds cobble-sized volcanic or carbonate rock and/or ceramic coarse aggregate (*caementa*). These clasts form a conglomeratic framework that reinforces the concrete. An early pozzolanic cementitious system, in which aggregates react with lime (*calyx*) hydrated with freshwater or seawater, binds the concrete. The components of this system then react with pore fluids to produce post-pozzolanic hydrated silicate phases, mainly strätlingite, Al-tobermorite, and phillipsite crystals, which remodel and toughen the concrete. A heterogeneous permeability structure facilitates these beneficial fluid-concrete interactions. The Ses Llumetes shipwreck provides a window into transport of *pulvis* pumiceous tephra as aggregate in marine structures. Roman natural scientists recorded accurate empirical observations

and hypotheses for these dynamic cementitious processes, which have great relevance to self-sustaining marine concrete infrastructure and alternatives to cement-based concrete.

- Advanced analytical methods validate Roman hypotheses for mechanisms of self-sustaining resilience in ancient architectural and marine concrete.
- An early pozzolanic system consolidates and strengthens the concrete, mainly through production of C-A-S-H (calcium-aluminum-silicate-hydrate) binding phase in a cementing matrix.
- Post-pozzolanic reactions with pore fluids produce silicate mineral cements and remodel the cementing matrix, toughening the concrete.
- A heterogeneous permeability structure derived from reactive aggregates, mainly volcanic tephra and cobble-sized *caementa*, aids these processes.
- Roman designs in modern marine concrete could produce beneficial interactions with saltwater, improving structural functionality and resilience.

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1. INTRODUCTION

In caementiciis autem structuris primum est de harena quaerendum, ut ea sit idonea ad materiem miscendam . . .

In concrete structures one must first inquire into the sand so that it will be suitable for mixing into mortar . . .

Vitruvius (1931, 2.4.1), ca. 25 BCE

The concrete of ancient Roman construction, fabricated through a vast expanse of Roman territories during the late Republican and Imperial eras, second century BCE to third century CE, has maintained cohesion and structural functionality for two millennia (Delaine 2024). A mortar mixture (*materia*) prepared with lime calcined from limestone in kilns at 850–900°C, reactive aggregate(s) mainly as volcanic tephra or ceramics, and freshwater or seawater binds the concrete, in which cobble-sized coarse aggregate clasts of hewn volcanic or carbonate rock and/or ceramic fragments (*caementa*) form a reinforcing framework (**Figure 1**). Marcus Vitruvius Pollio (ca. 90–20 BCE) described these materials in his treatise, *The Ten Books on Architecture*.

The binding processes in these aggregate structures, or *caementiciae structurae*, rely on fluid interactions with reactive aggregates to activate initial cohesion and sustain resilience at long ages. The reactive aggregates are principally scoriaceous volcanic tephra, or excavated sands (*harenae fossiciae*), for buildings in the Roman region and pumiceous volcanic tephra with a large fine-ash fraction, or dust (*pulvis*), for buildings surrounding the Gulf of Naples and massive marine concrete harbor installations (Vitruvius 1931, 2.4.1, 2.6.1–2.6.2, 5.12.2) (**Figures 1** and **2**). Enormous volumes of *pulvis* were eventually transported through the Mediterranean region to build concrete structures on land and in the sea (**Figure 3**).

By late first century BCE, builders had refined selections of volcanic tuff for dimension stone masonry and the coarse aggregate framework of concrete, as well as tephra aggregate for binding mortars. Roman natural scientists recorded empirical observations of these materials and stated hypotheses for early reaction processes and long-term resilience (*solidatis*) in *caementiciae structurae*, described in *Naturales quaestiones* (Seneca 1971, ca. 4 BCE–65 CE), *de Architectura* (Vitruvius 1931, ca. 90–20 BCE), and *Naturalis historia* (Pliny the Elder 1962, 23/24–79 CE). Long-term precipitation of silicate mineral cements toughens the concrete in bridges, cisterns, aqueducts, marine harbor structures and fish-raising tanks, and in architectural monuments, which have maintained

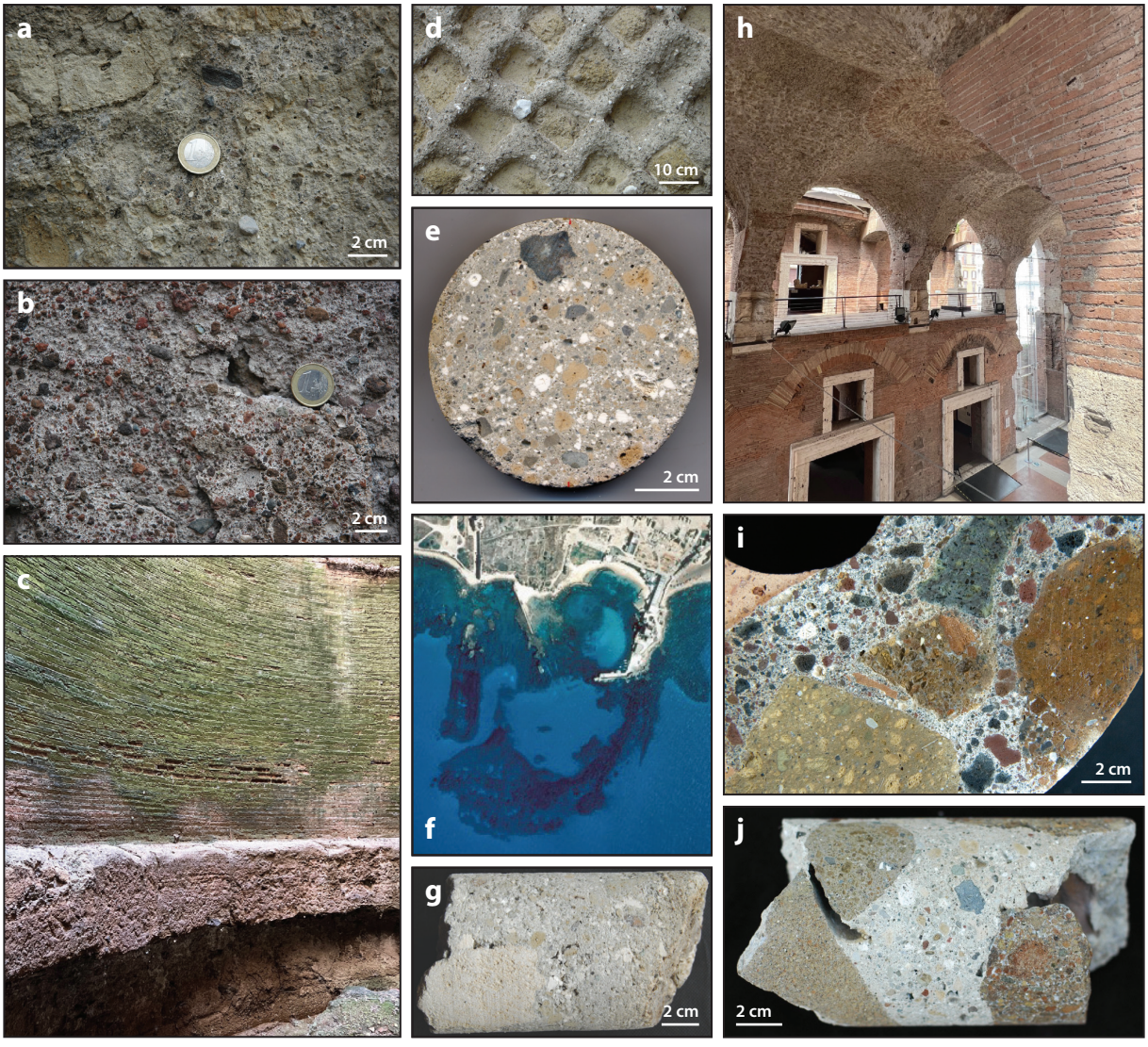


Figure 1

Photographs of Roman concrete structures, *caementiciae structurae*. (*a*) Temple of Concord, Rome, ca. 121 BCE, with *harena cana*, epiclastic volcanic sediments as mortar aggregate. (*b*) Temple of Concord, Rome, ca. 6 CE, with *harena rubra*, Pozzolane Rosse scoriaecous tephra, as mortar aggregate (Vitruvius 1931, 2.4.1). (*c*) Tomb of Caecilia Metella, Rome, ca. 30–10 BCE, moist internal wall of the 29-m-diameter brick-faced concrete drum and Tufo Lionato dimension stone reinforcement. (*d*) Outer wall of the *Piscina Mirabilis*, ca. 30 BCE, Miseno, *opus reticulatum* of Phlegrean tuff, eroded preferentially relative to mortar with Phlegrean pumiceous tephra, *pulvis*, as aggregate. (*e*) Drill core of marine concrete, *Baianus Sinus* breakwater, first century BCE, at Secca Fumosa with vitric and lithic fragments of Phlegrean pumiceous tephra as mortar aggregate. (*f*,*g*) *Caesarea Maritima* harbor, 23–15 BCE, and drill core of marine concrete, with Phlegrean pumice and tuff clasts. (*h*) Great Hall, Markets of Trajan, 96–115 CE, brick-faced wall concrete, travertine piers, and concrete cross-vaulted ceiling. (*i*) Drill core of conglomeratic wall concrete, Great Hall, with Pozzolane Rosse tephra as mortar aggregate. (*j*) Drill core of *Portus Traianus* concrete, ca. 115 CE. See **Figure 3** and **Supplemental Figure 1** for site locations. Credits: (*a*–*e*,*g*,*h*) M.D. Jackson, (*f*) J.P. Oleson, Caesarea Archaeological Harbor Excavation Project, and (*i*,*j*) C. Hagen.

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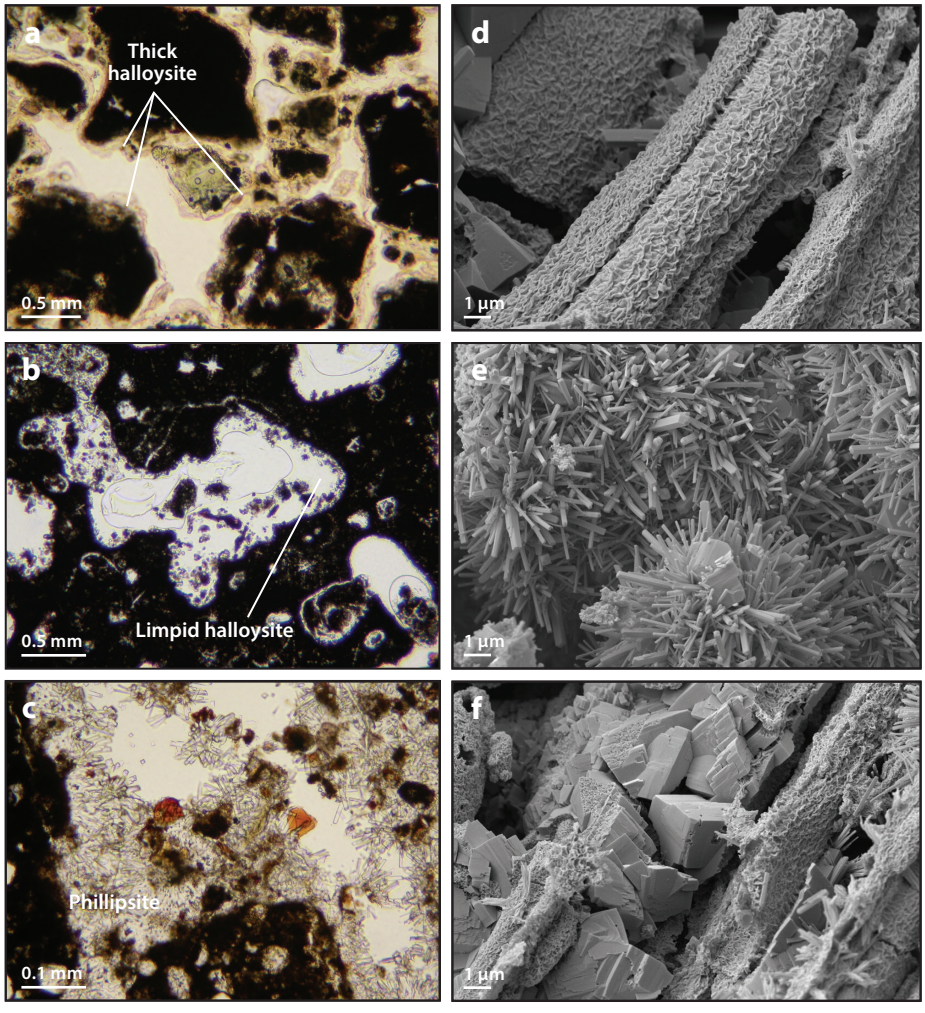


Figure 2

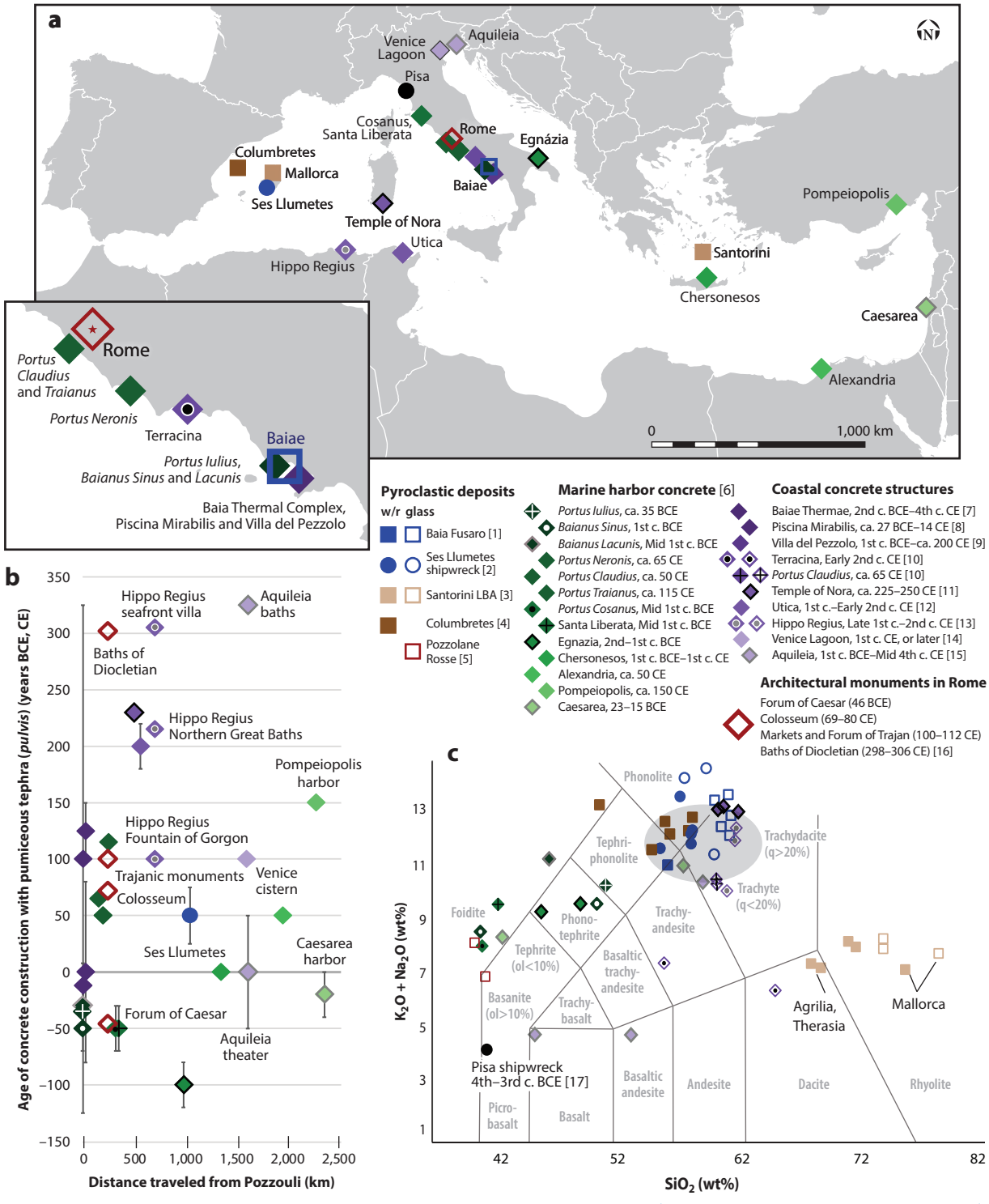
Reactive components of tephra deposits selected for mortar aggregate(s) from the volcanic landscapes of Rome and the Gulf of Naples (**Supplemental Figure 1**). Petrographic images of alteration facies from scoriaceous tephra of the Pozzolane Rosse pyroclastic flow, Colli Albani volcanic district, plane polarized light: (*a*) greatest, Castel di Leva quarry, sample 06-CASTEL-07; (*b*) intermediate, INGV drill core, sample INGV-08; and (*c*) least-altered, Santa Maria delle Mole drill core, 50.8 m. Scanning electron microscopy secondary election images of pumiceous tephra from Campi Flegrei pyroclastic deposits: (*d*) altered volcanic glass, (*e*) phillipsite crystals, and (*f*) chabazite crystals. Micrographs reprinted with permission from Rispoli et al. (2020). Abbreviation: INGV, Istituto Nazionale di Geofisica e Volcanologia.

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structural functionality for two millennia (e.g., Cantisani et al. 2002, Bianchi et al. 2011, Brandon et al. 2014, Jackson et al. 2014, Rispoli et al. 2020, Seymour et al. 2021, Medeghini et al. 2024).

The audacious concrete structures of Imperial age monuments in Rome—massive foundations, walls, and enormous vaulted ceilings—have fascinated scholars from the Renaissance through modern times. During the past 20 years, a cascade of discoveries has revealed precise sources of volcanic aggregates, installation procedures, fine-scale binding fabrics and mineral cements, additive mixtures, and initial insights into the physical properties of diverse concrete and dimension stone

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(*Caption for* ***Figure 3*** *appears on following page*)

Figure 3 (*Figure appears on preceding page*)

Marine and coastal concrete structures and shipwrecks with pumiceous tephra aggregate (*pulvis*) from Gulf of Naples pyroclastic deposits. (*a*,*b*) Map and diagram showing the age of concrete structures and estimated transport distance of Phlegrean tephra aggregate through the Mediterranean region, listed from north to south (**Supplemental Figure 2**, **Supplemental Table 1**). (*c*) Total alkali silica diagram (Le Bas et al. 1986), with tephra compositions plotted as w/r and glass analyses (**Supplemental Tables 2** and **3**, **Supplemental Geochemical Data**). For Terracina, Portus Claudius, and Hippo Regius, open symbols indicate pumice glass analyses. References: [1] Jackson et al. (2013a), [2] this study, [3] Druitt (2014), [4] Aparicio & Garcia (1995), [5] Marra et al. (2009), [6] Brandon et al. (2014), [7] Rispoli et al. (2019a), [8] Rispoli et al. (2020), [9] Rispoli et al. (2019b), [10] Marra et al. (2016a), [11] Dilaria et al. (2023a), [12] Fort et al. (2024), [13] Djerad et al. (2022), [14] Dilaria et al. (2024), [15] Dilaria et al. (2022, 2023b), [16] Marra et al. (2013), and [17] Pecchioni et al. (2007) and Marra & D'Ambrosio (2013). Abbreviations: LBA, Late Bronze Age eruption; w/r: whole rock.

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structures (e.g., Jackson & Marra 2006, Hobbs & Siddal 2011, Brandon et al. 2014, Artioli et al. 2019, Lancaster 2019, Delaine 2024). Other studies have translated the geotechnical expertise of Roman builders to fabricating replicas and analogs of the ancient mortars and concretes, assessing chemical and mechanical resilience of these materials, and comparing their performance specifications to those of modern Portland cement concretes (Samuelli Ferretti 1995, Goldsworthy & Zhu 2009, Brandon et al. 2014, Jackson et al. 2014).

Portland cement is made by sintering limestone and aluminosilicate materials at 1,450°C to produce a quenched product, clinker, which is pulverized to fine powder with a calcium sulfate addition. The chemical reaction of this powder with water (Marchon & Flatt 2016) produces cohesion and strength in the cement paste that binds inert sand- and gravel-sized aggregate in modern concrete. The English word cement comes from the Old French *ciment*, derived from the Latin *caementum*, or hewn stone, perhaps referring to limestone as a source for lime (CaO). The English word concrete comes from the Latin *concretum*, meaning solidified, compact, or condensed, derived from *concrescere*, to grow together [*con-* (together) and *crescere* (to grow)]. No direct equivalents of the English terms cement and concrete, however, appear in Roman texts. Furthermore, no cement product was employed in the ancient structures.

This review integrates ancient texts, archaeological observations, and analytical descriptions of the geochemical, mineralogical, and material characteristics of well-preserved Roman concrete structures to provide new insights into the relevance of *caementiciae structurae* to modern concrete construction. Several fundamental questions guide ongoing research:

1. Through what skill (*fabrica*) and reasoning (*ratiocinatio*) did Roman architects and engineers create the knowledge (*scientia*) to transform volcanic tephra and tuff into concrete structures, *caementiciae structurae*, that form marvels of architectural brilliance and sustainable engineering practice?
2. What hypotheses did Roman natural scientists advance for how pozzolanic cementitious processes requiring large initial energy inputs—calcination of lime and explosive pyroclastic eruptions—transform to self-sustaining cementitious processes in concretes that "maintain themselves … into very old age" (*ad summam vetustatem*) (Vitruvius 1931, 2.8.6)? How does the transport of pumiceous tephra (*pulvis*) from the Gulf of Naples through the Mediterranean region reflect Roman builders' adherence to these guiding principles?
3. What do the permeability structures of the ancient architectural and marine concretes reveal about the chemical and mechanical resilience that Roman builders sought to develop in different types of structures?
4. How have the concrete structures sustained themselves to retain cohesion and structural functionality over two millennia in response to floods, earthquakes, and seawater ingress—the environmental threats that cause the often-rapid demise of modern

cement-based concretes? How can these principles be applied to concrete of the twenty-first century?

2. GEOLOGIC COMPONENTS: *MATERIA*, *CAEMENTA*, *CALYX*

Ancient Roman concrete contains approximately 70–75 vol% *caementa* in terrestrial structures and 40–45 vol% *caementa* in marine structures, on average; the binding mortar (*materia*) has an approximately 1:2.7 vol% ratio of hydrated lime to reactive aggregate (Samuelli Ferretti 1995, Brandon et al. 2014, Delaine 2021). Rock (or ceramic) aggregates thus constitute 85–90 vol% of the ancient structures. A stiff mix was packed into forms, not poured, with *caementa* compacted in the mortar (Oleson et al. 2006). This produces a clast-supported, conglomeratic concrete fabric (**Figure 1*g*,*i*,*j***) commonly hidden from view by *incertum* and *reticulatum* facings during the late Republican era (**Figure 1*d***) and brick facings, *testaceum*, mainly during the Imperial era (**Figure 1*c*,*b***). There is great diversity in formwork for construction ranging from immense vaulted ceilings in architectural structures to reef-like marine structures in the sea (Brandon et al. 2014, Lancaster 2015).

2.1. A Long Fascination with Roman Architectural Concrete

Architects and natural scientists of the Renaissance such as Leon Battista Alberti [1988 (1452)], the Age of Enlightenment such as Barthélémy Faujas de Saint-Fond (1780), and the nineteenth century including Faustino Corsi Romano [1991 (1828)] and August Choisy (1873) described Roman building stones and concrete, including diverse sources of volcanic pozzolan by Faujas de Saint-Fond (1780) (Thiéry 2024). Fascination with Roman construction materials grew during the late nineteenth and early twentieth centuries as scholars explored monumental architecture and ruins in Rome and the volcanic rocks of its surroundings (**Supplemental Figure 1*a***). In research supported by the Carnegie Institution, H.S. Washington (1906) employed geochemical and mineralogical analyses to describe the potassic nature of the Roman Comagmatic Province, and E.B. Van Deman (1912a,b) made systematic, empirical descriptions of the aggregate components of the architectural concrete. T. Frank (1924) inferred the provenance of Republican era construction materials. Further hypotheses for sources of these materials and building technologies were developed by G. Lugli (1957) and M.E. Blake (1947, 1959, 1973), yet they were not integrated with contemporaneous studies of the stratigraphy of the Monti Sabatini and Colli Albani volcanic districts (Scherillo 1944–1946, Fornaseri et al. 1963).

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The publication of *La geologia di Roma*, *Il centro storico* (Funiciello 1995) created a stratigraphic, hydrologic, seismologic, and geotechnical foundation for identifying sources of the construction materials of the ancient monuments and their responses to an exceptionally dynamic geologic environment. Studies of the Quaternary fluvial-volcanic stratigraphy and geochronology of the Capitoline Hill distinguished pyroclastic deposits filling separate paleovalleys; these deposits record the responses of the Tiber River to sea level change in the nearby Tyrrhenian Sea (Alvarez et al. 1996). $^{40}Ar/^{39}Ar$ age dating of the potassic volcanic deposits led to a formalized volcanic stratigraphy calibrated to sea level rise and fall over the past 800 ka (Karner et al. 2001) (see the sidebar titled Volcaniclastic Rock in Ancient and Modern Concrete).

2.2. Volcanic Foundations of the Ancient Concrete

Roman concrete construction has its roots in explosive pyroclastic eruptions of the Monti Sabatini and Colli Albani volcanic districts, which underlay Rome and its surroundings, and the Campi Flegrei and Somma Vesuvius volcanic districts, which surround the Gulf of Naples (Italy)

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VOLCANICLASTIC ROCK IN ANCIENT AND MODERN CONCRETE

Tephra and tuff are pyroclastic deposits produced by explosive eruptions (*pyro* = fire + *clastic* = broken in pieces). Tephra is composed of fragments: glass quenched from molten rock, crystals, and host rocks. Tuff is the rock produced by lithification of tephra, mainly through growth of mineral cements. Tufa, or travertine, is calcium carbonate rock deposited by thermal springs and lakes. These deposits occur throughout the Roman region, and they surround the Gulf of Naples.

Romans used tephra in its original particle size distribution as mortar aggregate. The coarse ash- and lapilli-sized fractions, 1–6.4 mm, form reactive aggregate in the mortar. The fine-ash-sized fraction, less than 1 mm, reacted with hydrated lime and forms part of a complex cementing matrix (**Figure 4**). In modern concrete, volcanic rock pozzolan is ground to less than 100 μm to replace cement. The aggregate is inert sand and gravel, 0.5–4.75 mm.

Conventional concretes with volcanic pozzolan develop good strength at 28–90 days of hydration, 25–35 MPa. They also show resistance to chemical attack. In marine structures, however, a stiff fracture response produces cracks and seawater ingress that corrodes steel reinforcement. A Roman analog marine concrete may be a self-sustaining, highly resilient alternative.

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(**Supplemental Figure 1**). Romans quarried specific pyroclastic deposits from the complex stratigraphies of these volcanic districts for use as dimension stone or as concrete aggregates.

Pyroclastic deposits undergo alteration when fluids in pores are modified through reactions with volcanic components. Opal and poorly crystalline clay mineral surface coatings, or textures, may form in tephra of volcanic soils, for example (Stoops et al. 2018). The uppermost horizon of a voluminous mid-Pleistocene pyroclastic flow deposit, Pozzolane Rosse, which erupted from the Colli Albani volcanic district at 456 ± 3 ka (Marra et al. 2009), records a hydrolytic ultisol that formed during a humid period preceding marine isotope stage 11 (Jackson et al. 2010); thick, opaque halloysite clay mineral coatings surround dusky yellow-brown scoria clasts (**Figure 2*a***). An intermediate alteration facies has delicate opal and translucent halloysite surface coatings, however; leucite, a potassic silicate ($KAlSi_2O_6$), is commonly replaced by analcime, a sodic zeolite. A least-altered facies at the base of the deposit also has phillipsite and chabazite zeolites, and leucite is mainly preserved (**Figure 2*c*,*d***). The dusky red to dark gray tephra of these two Pozzolane Rosse alteration facies are the excavated sands described by Vitruvius (1931, 2.4.1), *harena rubra* and *harena nera*. These facies form the principal mortar aggregate of concrete in monuments constructed in Rome from the very late Republican through Imperial eras; quarries occur through the volcanic landscape surrounding Rome (Van Deman 1912a,b; Jackson & Marra 2006; Jackson et al. 2010, 2014; Marra et al. 2011; Belfiore et al. 2015a). Fresh glass has a potassic foidite composition (**Figure 3**), yet whole rock compositions show wide variations resulting from alteration processes (Marra et al. 2009, 2011; Belfiore et al. 2015a).

Zeolite minerals crystallize in tephra deposits when the cation to hydrogen ion ratio and other ionic activities in fluids are relatively high; the solutions become more alkaline and enriched in Na, K, Ca, and Si as they travel through the deposit (Sheppard & Hay 2001). In Campi Flegrei deposits (**Supplemental Figure 1*b***), post-eruptive hydrolysis and dissolution of glass, phonolitic to trachytic in composition (**Figure 3*c***), in the presence of condensed water vapor generate alkaline fluids from which zeolites precipitate (de'Gennaro et al. 1999, 2000). **Figure 2*d–f*** shows altered volcanic glass and typical phillipsite and chabazite zeolites (Rispoli et al. 2020). Airfall deposits from tephra clouds are less zeolitized than hydromagmatic deposits that retain pore water and water vapor, grow zeolite mineral cements, and lithify to form tuff. By first century BCE, Roman builders systematically selected pumiceous tephra, *pulvis*, quarried "in the regions of Baie and

Mount Vesuvius that naturally effects an admirable result" for the mortar of maritime harbor concrete; zeolitized tuff served as coarse aggregate (*caementa*) (Vitruvius 1931, 2.6.1).

Early and mid-Republican era monuments in Rome mainly utilized dimension stone construction with local tuffs that have low strength and durability, described as soft and yielding (*molles*) (Vitruvius 1931, 2.7.1; Jackson et al. 2005, Jackson & Marra 2006). Although Roman builders protected the tuff masonry with stucco or plaster coatings (Vitruvius 1931, 2.7.2, 7.5.8), innovative architectural designs in the later Republican era (Delaine 2024) required a more robust construction material that thrived in humid environments. The development of *caementiciae structurae* is thought to have evolved at several sites within the Italian peninsula during early second century BCE: in domestic architecture at mid-second century BCE and in public architecture possibly only from 130 to 120 BCE (Mogetta 2021). [Scholars commonly refer to this concrete as *opus caementicium*, from the *caementa* coarse aggregate, but this term occurs only once in primary sources, as a second-century CE sanctuary inscription in today's northeastern Greece, *CIL* 3.633 (Mogetta 2021, p. 10).]

Aggregates of some Republican era mortars were excavated from epigenetic alluvial deposits left by the Tiber River over a series of aggradation sequences. Other aggregates were quarried from tunnels through tephra and tuff pyroclastic flow deposits (Marra et al. 2015, 2016a,b). The mortar aggregate of the Temple of Concordia in Rome, 121 BCE, for example, is composed of volcaniclastic sediments excavated from the building site at the base of the Capitoline Hill (**Figure 1*a***); a reconstruction in 10 CE used Pozzolane Rosse aggregate from the intermediate alteration facies (Van Deman 1912a, Jackson et al. 2007) (**Figure 1*b***).

By mid-first century BCE, builders had developed a good empirical knowledge of the diverse material properties of tuffs in the Roman region (Nappi et al. 1979, De Casa et al. 1999, Jackson et al. 2005). Tuff extracted from the quarry for use as dimension stone (*quadratis lapidibus*) or concrete structures (*caementiciae structurae*) should weather in place for two years before being selected as sufficiently durable to be used in aboveground structures or, conversely, to be "thrown in with the foundations" (Vitruvius 1931, 2.7.5). Scholars commonly translate *caementa* as rubble, mainly implying quarried stone (Mogetta 2021). In the Imperial era and later antiquity, *caementa* also includes stone, brick, mortar, and concrete materials derived from demolition or destruction of buildings or structures (e.g., Underwood 2019). Further research is needed to clarify the use of recycled materials as aggregates in *caementiciae structurae*.

2.3. A Hydrated Lime Activator

Volcanic glass, zeolite minerals, opal, and poorly crystalline clay minerals are natural pozzolans, derived from the modern city of Pozzuoli, ancient Puteoli, a maritime transport hub for *pulvis* (**Figure 3*a***; **Supplemental Figure 1*b***). Pozzolans react with hydrated lime, or calcium hydroxide [$Ca(OH)_2$], to produce durable cementing hydrates (Massazza 1998, Mertens et al. 2009, Elsen et al. 2022). In Roman concrete, the volcanic glass and mineral components of tephra (**Figure 2**) react pozzolanically with hydrated lime to produce a poorly crystalline calcium-aluminum-silicate-hydrate (C-A-S-H) binding phase that bonds tephra particles within a complex mortar (**Figure 4**). Myriad methods for quicklime hydration and mixing with reactive aggregates (Traini 2013, Copsey 2019) are recorded by the microstructural features of Roman mortars, demonstrated in part through mortar and concrete replicas and reproductions (e.g., Oleson et al. 2006, Goldsworthy & Zhu 2009, Brandon et al. 2014, Jackson et al. 2014). Silicate cementitious hydrates, rather than calcium carbonate phases, dominate mortars accessed by drill cores in thick walls or underwater structures (e.g., Jackson et al. 2013b, 2014, 2017; Brandon et al. 2014, pp. 166–75; Rispoli et al. 2020).

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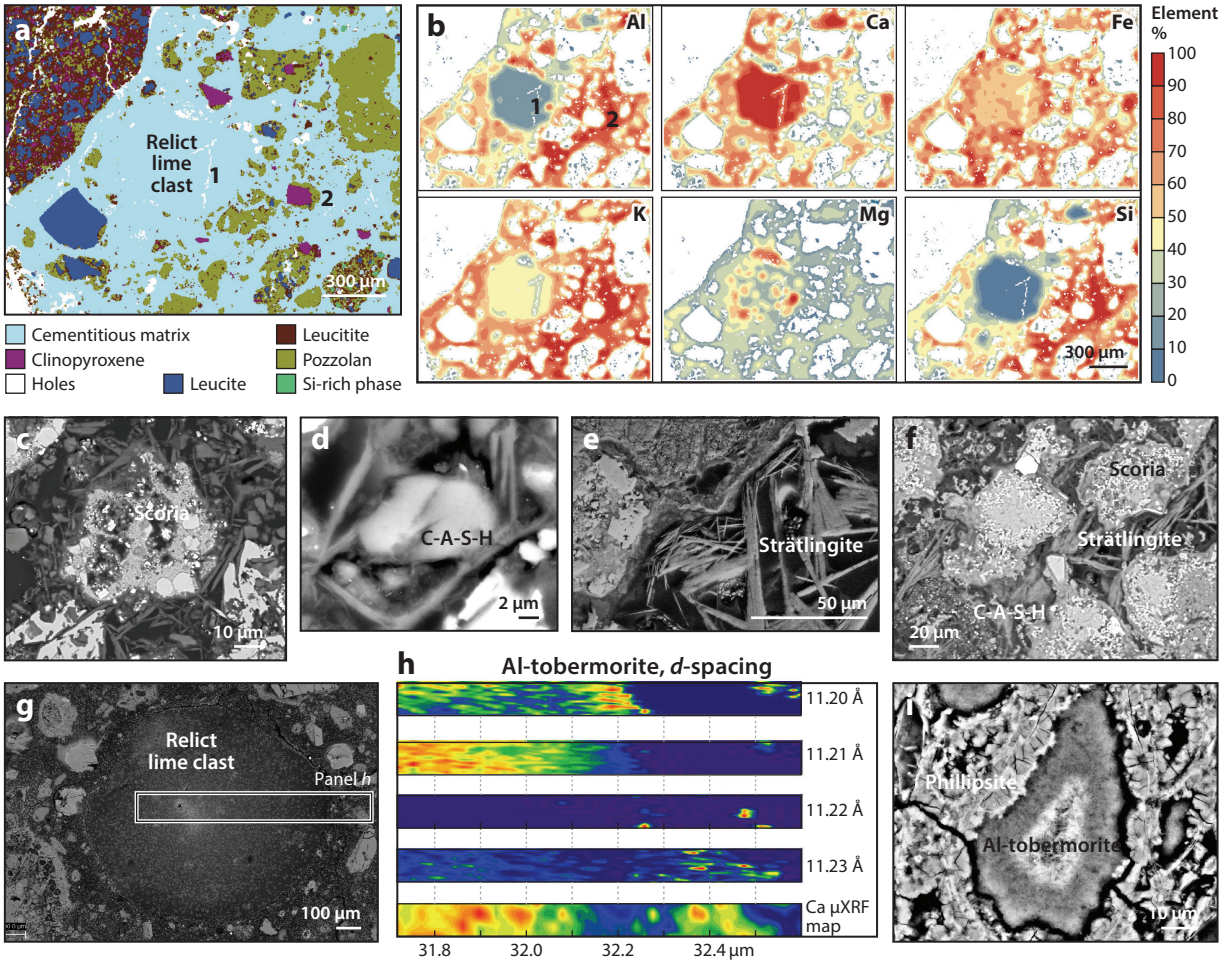


Figure 4

Analytical maps and microscopy images of the cementing matrix of ancient Roman mortars. Villa dei Quintili, Rome, mid-second century CE: (*a*) mortar components and (*b*) kernel density maps of spectroscopic measurements of Al, Ca, Fe, K, Mg, and Si. Reprinted with permission from Belfiore et al. (2015b). Post-pozzolanic cementitious phases, SEM-BSE images: (*c*,*d*) beneficial destabilization of C-A-S-H binding phase, Tomb of Caecilia Metella (ca. 30–10 BCE) (**Figure 1*c***); (*e*,*f*) strätlingite crystals, Theater of Marcellus (44–11 BCE) (after Jackson et al. 2011) and Grande Aula, Markets of Trajan (96–115 BCE) (**Figure 1*b*,*i***). *Baianus Sinus*, Bay of Pozzuoli, first century BCE (**Figure 1*e***): (*g*,*h*) SEM-BSE image, calcium µXRF map and *d*-spacing µXRD maps of Al-tobermorite interlayer *d*-spacings in a relict lime clast and the cementing matrix (after Jackson et al. 2018). (*i*) SEM-BSE image, post-pozzolanic Al-tobermorite crystals in the vesicle of a pumice clast surrounded by Campi Flegrei phillipsite crystals (**Figure 2*e***). See **Supplemental Figure 3** for Al-tobermorite and phillipsite compositions. Abbreviations: BSE, backscattered electron; C-A-S-H, calcium-aluminum-silicate-hydrate binding phase; SEM, scanning electron microscopy; µXRD, X-ray microdiffraction; µXRF, X-ray microfluorescence.

3. CEMENTITIOUS FABRICS OF THE ANCIENT CONCRETES

3.1. Highly Refined Concretes in Monumental Architecture

Supplemental Material >

A matrix composed of a poorly crystalline, hydrated calcium-(aluminum)-silicate cementitious phase [C-(A)-S-H], and partially reacted tephra of the Pozzolane Rosse pyroclastic flow, in a broad particle size distribution (Columbu et al. 2018), forms the quintessential binding fabric of the

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architectural mortars. Novel analytical methods describe the chemical and mineralogical characteristics of this matrix in a bedding mortar from Villa dei Quintili, second century CE, on the Via Appia south of Rome (**Supplemental Figure 1*a***). Multivariate statistical analysis of scanning electron microscopy electron dispersive spectroscopy maps computes the element distribution among and within specific phases (Belfiore et al. 2015b) (**Figure 4*a*,*b***). Calcium-rich binding phase occupies relict clasts of hydrated lime (site 1) yet is enriched in silicon, aluminum, and potassium where it surrounds relicts of fine-ash-sized tephra (site 2). The analyses indicate highly heterogeneous compositions at micrometer scale. Trace element analyses of underburned limestone clasts reveal that quicklime was calcined from Early Jurassic limestone, Calcare Maiolica, in the nearby Monti Cornicolani (Fichera et al. 2015) (**Supplemental Figure 1*a***).

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More than 200 years earlier, builders perfected fabrication of this complex mortar. In the indurated concrete walls of the Tomb of Caecilia Metella, ca. 30–10 BCE, frequently saturated with groundwater (and rainwater) (**Figure 1*c***), C-A-S-H binding phase is reorganized into halos and tendril-like strands or split into elongate features (**Figure 4*c*,*d***). The C-A-S-H incorporated Al^{3+} and K^{+} from pore fluids derived from dissolution of leucite crystals in the least-altered facies of Pozzolane Rosse tephra aggregate (**Figure 2*c***). The large potassium cations appear to have induced a structural destabilization in the layered C-A-S-H structure, yet these changes seem to have enhanced bonding in the concrete (Seymour et al. 2021). In the concrete substructure of the Theater of Marcellus, 44–11 BCE, repeatedly saturated with floodwaters from the Tiber River (Aldrete 2007, Jackson et al. 2011), acicular and platey crystals of strätlingite [$Ca_2Al_2SiO_2(OH)_{10}\cdot 2.25H_2O$] reinforce interfacial zones of Pozzolane Rosse scoriae from the intermediate alteration facies (**Figures 2*b*** and **4*e***).

Strätlingite crystals also reinforce the mortar of the wall concrete of the Great Hall of the Markets of Trajan (96–115 CE) (Vitti 2007) (**Figures 1*b*,*i*** and **4*f***). A precise replica of this mortar crystallizes strätlingite through chemical and mineralogical processes at 90 days of hydration, after pozzolanic consumption of hydrated lime, or portlandite [calcium hydroxide, $Ca(OH)_2$], is complete (Jackson et al. 2014). Indeed, strätlingite is not compatible with portlandite and decomposes in its presence (Snellings et al. 2012, Okoronkwo & Glasser 2016).

An innovative arc-bending test investigates the chemical and mechanical resilience of the wall mortar replica (Brune et al. 2013, Jackson et al. 2014). In the applied force–crack mouth opening displacement (CMOD) diagram (**Figure 5*a***), curves representing experimental (solid) and computational (dashed) results for tests at 28, 90, and 180 days of hydration indicate an early region (labeled 1) that signals growth of microcracks before the curve reaches its peak (labeled 2). A smooth post-peak descent (labeled 3) indicates stable growth of a critical macrocrack and weakening of the specimen (labeled 4), similar to fracture of quasi-brittle materials (Bažant et al. 2009) in cement-based concrete (van Mier 1997, Bažant & Planas 1998). A great deal more work, measured as fracture energy (G_F), is required to propagate microcracks at 90 and 180 days of hydration (**Figure 5*b***). Strätlingite crystals along interfacial zones of reactive scoriae (**Figure 5*c***) increase fracture toughness, the resistance to crack propagation and unstable fracture. The architectural concrete gained mechanical resilience through these post-pozzolanic processes very early in its long history of service life.

3.2. Building Concrete Structures in the Sea

When builders packed *caementa* and pumiceous tephra-hydrated lime mortar in submarine formwork (**Figure 1*e–g*,*j***), they created a pozzolanic system at pH $\geq$ 12, buffered by hydrated lime, which produced three principal cementitious phases. C-A-S-H bonded a complex cementing matrix. Layered double hydroxide crystals captured chlorine and sulfate anions in seawater

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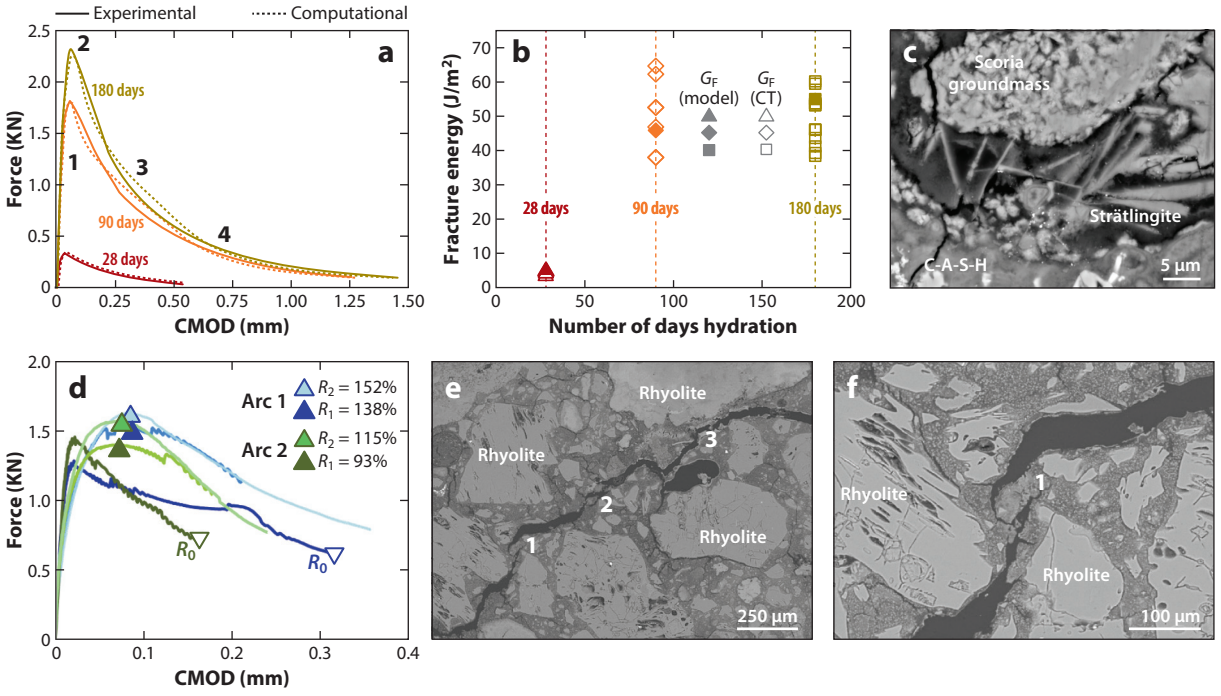


Figure 5

Fracture processes recorded by arc-bending tests. Markets of Trajan wall mortar replica with Pozzolane Rosse tephra aggregate hydrated in freshwater: (*a*) experimental (*solid*) and computational (*dashed*) force versus CMOD curves; (*b*) fracture energy (G_F) measured from mapped fractures and compared with G_F determined through finite element analysis (Brune et al. 2013); and (*c*) SEM-BSE image, C-A-S-H binding phase and strätlingite crystals, 180 days of hydration. Roman analog marine concrete with rhyolite aggregate hydrated in synthetic seawater: (*d*) force versus CMOD responses for the baseline arc-bending test at 46 days of hydration, R_0, the first, R_1, and second, R_2, repeat tests at 433 and 1,056 days of hydration, showing ILR = $(F_{max}{}^{Ri} - F_{stop}{}^{0})/(F_{max}{}^{0} - F_{stop}{}^{0})$ (Ferrara et al. 2014). SEM-BSE images, R_2: (*e*) bridging microstructure, en echelon microcrack segments and (*f*) C-A-S-H infill in tortuous microcrack segment. Abbreviations: BSE, backscattered electron; C-A-S-H, calcium-aluminum-silicate-hydrate binding phase; CMOD, crack mouth opening displacement; CT, computed tomography; ILR, index of load recovery; SEM, scanning electron microscopy.

hydrating fluid (Zhai et al. 2022); these are mainly calcium-chloroaluminate crystals, hydrocalumite, and calcium-sulfoaluminate crystals, ettringite (Jackson et al. 2012, 2024a). Al-tobermorite, an unusual, layered calcium-silicate-hydrate mineral, $[Ca_4(Si_{5.5}Al_{0.5}O_{17}H_2)]Ca_{0.2}{\cdot}Na_{0.1}{\cdot}4H_2O$ (Taylor 1992), with 11 Å interlayer *d*-spacing and cation-exchange properties for heavy metals (e.g., Komarneni et al. 1987), crystallized in hydrated lime clasts (Stanislao et al. 2011, Vola et al. 2011, Jackson et al. 2013a) (**Figure 4*g*,*h***). In an experimental concrete *pila* reproduction that used *pulvis* pumiceous tephra as highly reactive pozzolan (Sersale & Orsini 1969) (**Figure 2*d–f***), portlandite remained present at four years of hydration (Oleson et al. 2006; Gotti et al. 2008; Brandon et al. 2014, pp. 174–80). At 19 years, portlandite is fully consumed and the C-(A)-S-H binding phase continues to incorporate silica (Sohounme et al. 2026). Pozzolanic reaction in larger Roman harbor structures was probably largely complete within 10–20 years.

3.2.1. Resilience of the young marine concrete. The arc-bending test also provides insights into the resilience of the early-age marine concrete. **Figure 5*d*** shows the force-CMOD diagram for iterative arc-bending tests of a reproduction of the marine concrete with glassy rhyolite reactive aggregate, hydrated in synthetic seawater. After a baseline test at 46 days of hydration (R_0) followed by subsequent tests after 387-day and 623-day recovery intervals (R_1, R_2), the arc specimens gain

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significant mechanical carrying capacity; this is a cumulative increase of approximately 50% above the baseline carrying capacity (M.D. Jackson et al., unpublished article). En echelon microfractures 200–300 μm in length link through bridging microstructures (**Figure 5*e*,*f***), indicating a strain-tolerant material fabric. At nearly three years of hydration, the concrete remains in a chemical state of active pozzolanic reaction and production of C-A-S-H binding phase. Recovery of carrying capacity between damage events prevents the progressive increase in total degradation toward failure that describes many Portland cement concrete systems (Torres-Acosta et al. 2007, Ferrara et al. 2014).

3.2.2. Crystallization of mineral cements. The intricate fabrics of mineral cements in the marine concrete have been illustrated by fine-scale X-ray microfluorescence and X-ray microdiffraction analyses using a synchrotron monochromatic beam (Jackson et al. 2024a). For example, the precise interlayer *d*-spacings of Al-tobermorite crystals, 11.20–11.23 Å, are mapped in a relict clast of hydrated lime and the cementing matrix (**Figure 4*g*,*h***) and then correlated with other material characteristics (Jackson et al. 2013a,b, 2018). Al-tobermorite that crystallizes in pumice clasts contains more silica, alumina, and alkalis (**Figure 4*h***; **Supplemental Figure 3**).

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Phillipsite also crystallizes within the cementing matrix (Stanislao et al. 2011; Vola et al. 2011; Jackson et al. 2012, 2017); these crystals contain less silica than phillipsite of Campi Flegrei pumiceous tephra (**Figure 2*e***; **Supplemental Figure 3**). They indicate a decrease in the pH of pore fluids from the portlandite buffered system, pH ≥ 12, to the low-temperature phillipsite stability field, pH 8–10 (de'Gennaro et al. 2000, Sheppard & Hay 2001). The compositions of phillipsite and silica-enriched Al-tobermorite crystals are similar to mineral cements that crystallize in young oceanic basaltic tephra at Surtsey volcano, Iceland (Jakobsson & Moore 1986, Montesano et al. 2023, Jackson et al. 2024b) (**Supplemental Figure 3**). They indicate that the chemical and mineralogical system evolved from pozzolanic reactions with hydrated lime to post-pozzolanic reactions with alkaline pore fluids, produced by slow infiltration and reaction of seawater within the massive concrete structures (Jackson et al. 2017).

3.3. Porosity and Permeability Fabrics

The architectural and marine concretes have highly heterogeneous permeability structures (**Figure 6**; **Supplemental Table 4**), shown by maps of in situ permeability measurements (Farquharson et al. 2015) on the surfaces of drill cores from the walls of the Great Hall, the Markets of Trajan, and a mole at *Portus Traianus*, 115 CE, at the mouth of the Tiber River at Portus (Vitti 2007; Brandon et al. 2014, pp. 256–58) (**Figures 1*b–j***, **3*a***, and **6*a*,*b***). The cementing matrix of the architectural mortar with Pozzolane Rosse aggregate and the marine mortar with a mixture of Pozzolane Rosse and pumiceous tephra, *pulvis*, aggregate average 1.56×10^{-15} m^2 and 3.12×10^{-15} m^2, respectively (**Figure 6*c*,*d***). Interfacial transition zones along the tuff and ceramic *caementa* show large permeability variations, however, from 10^{-14} m^2 in bonded contacts to 10^{-12} m^2 in porous contacts. Strength, measured as point source load tests, also varies along these zones (Jackson et al. 2009, Columbu et al. 2018).

The connected porosity and permeability of subcores of the ancient mortars are substantially higher than those of high-performance blended Portland cement concrete, which has low connected porosity, 10%, and very low permeability, 3×10^{-18} m^2 (**Figure 6*e***; **Supplemental Table 4**). These elevated values do not seem to harm the ancient concrete, however. Although the marine mortars have high porosity, mainly 45–55%, pore radii are very small, 4–120 nm, an order of magnitude less than cement-based mortars but similar to Phlegrean tuff (Gotti et al. 2008; Brandon et al. 2014, pp. 180–83; Rispoli et al. 2020). The vesicular structure and reactivity of the tephra aggregates seem to enhance slow permeation of fluids, which evolve

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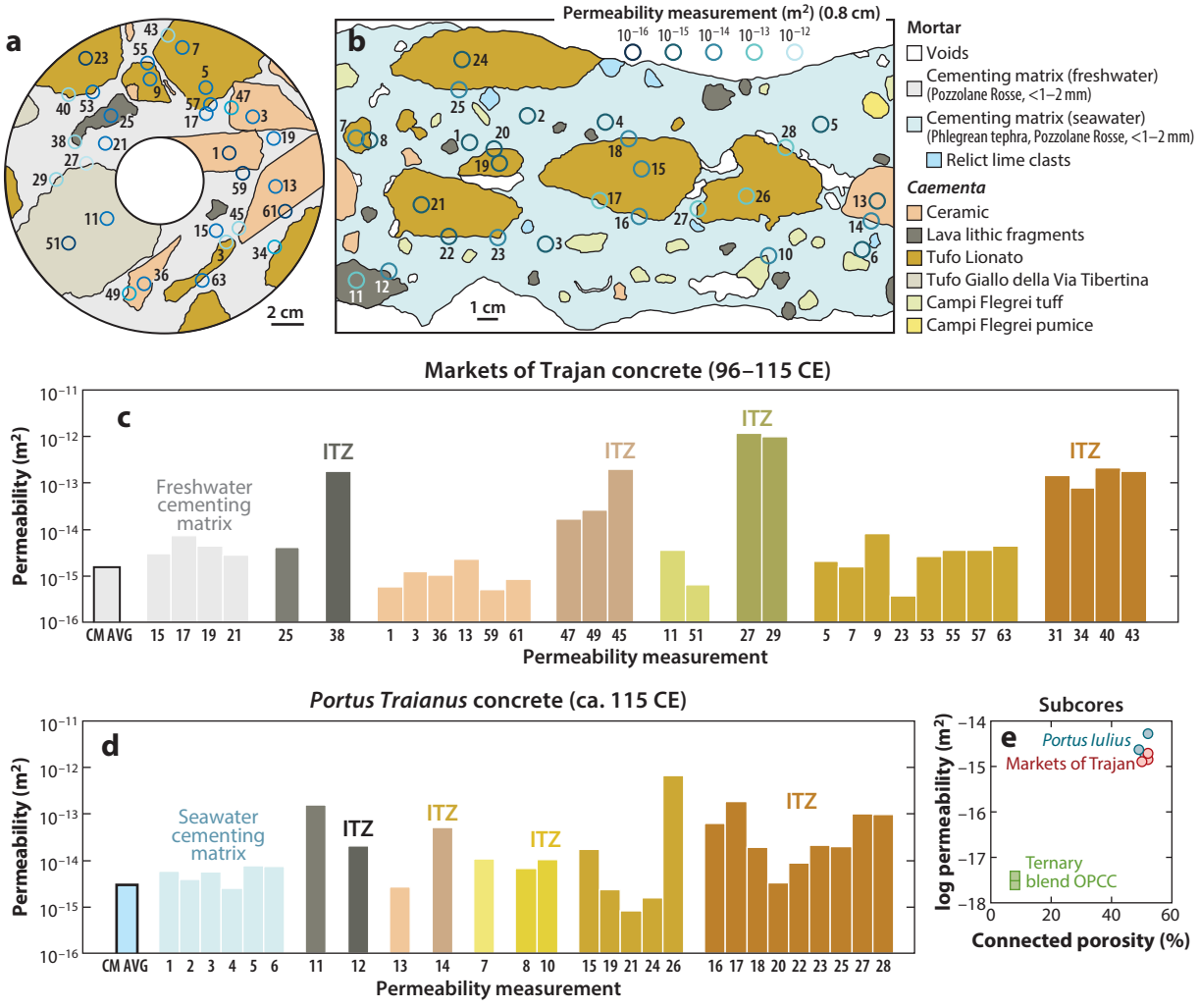


Figure 6

The permeability structure of Trajanic era architectural and marine concretes. (*a*,*b*) Maps of in situ permeability measurements, slice of Markets of Trajan wall concrete drill core (20 cm diameter), and the surface of *Portus Traianus* drill core (9 cm diameter). (*c*,*d*) Permeability measurements of concrete components and their interfacial transition zones with the cementing matrix, over an area of 0.5 cm^2, using a handheld permeability meter with 0.8-cm-diameter nozzle. (*e*) Permeability as a function of connected porosity for 2 × 4 cm subcores of ancient mortars and ternary blended OPCC, measured using a benchtop gas permeameter at a confining pressure of 1 MPa. See **Supplemental Table 4** for data and analytical methods. Abbreviations: ITZ, Interfacial Transition Zone; OPCC, ordinary Portland cement concrete.

in composition to precipitate strätlingite, Al-tobermorite, and phillipsite mineral cements that toughen the cementing matrix and further slow fluid flow.

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4. RESOURCE PROVENANCE AND TRANSPORT

Pumiceous tephra from deposits surrounding the Gulf of Naples has been identified in marine and coastal concrete structures, first century BCE to third century CE, through the Mediterranean

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region (**Figure 3**). The central distribution site seems to have been Puteoli, now Pozzuoli (Strabo 1917, 5.4.6).

4.1. A Diaspora of Pumiceous Tephra, *Pulvis*

Chemical analyses of pumice clasts in the mortars of the marine concretes provide clues to pyroclastic source deposits. Trachytic glass, primary volcanic crystals, mainly sanidine, and authigenic minerals, mainly phillipsite, are reactive components in the ancient mortars (**Figure 2*d*,*e***). Pumice clasts separated from the mortars should therefore record depletion in major elements relative to source compositions. Indeed, compositions of pumice clasts from drill cores of concrete harbor structures (**Figure 3*c***) show lower SiO_2 and $Na_2O + K_2O$ in the total alkali-silica diagram, relative to compositions from Campi Flegrei deposits (Brandon et al. 2014); these are elements involved in authigenic precipitation of phillipsite, for example (**Figure 2*e***). Rare earth elements (REE) are thought to remain immobile in glass and crystals. When analyses of REE in pumice clasts separated from the marine mortars are compared with the same elements in Campi Flegrei pyroclastic deposits and coupled with mineralogical data, a close correlation is shown for the harbor concrete structures (Brandon et al. 2014, pp. 148–59).

Pumiceous tephra aggregate occurs in Republican era concrete structures less than 5 km from deposits near Baia (**Figure 3*b***): the Thermal Complex of Baia, second century BCE, *Piscina Mirabilis*, ca. 27 BCE–14 CE (**Figure 1*d***), and harbor structures at *Baianus Lacunis*, *Baianus Sinus* (**Figure 1*e***), first century BCE, and *Portus Iulius*, 35 BCE (Brandon et al. 2014, pp. 260–65; Rispoli et al. 2019a,b, 2020). Approximately 20,000 m^3 of pumiceous tephra and tuff *caementa* were transported ~2,400 km to *Caesarea Maritima* at Sebastos, Israel, ca. 23–15 BCE (**Figure 1*f*,*g***). Tens of thousands of kilos of pumiceous tephra were also shipped to marine structures along the west coast of the Italian peninsula (**Figure 3*b*,*c***; **Supplemental Figure 2**, **Supplemental Table 1**). The vaulted concrete architecture of monuments in Rome also contains pumiceous tephra from the Gulf of Naples volcanic districts (Marra et al. 2013), as do coastal structures in Sardinia, northern Africa, Aquilea, and the lagoon of Venice (Djerad et al. 2022; Dilaria et al. 2023a,b, 2024; Fort et al. 2024).

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4.2. Ses Llumetes Shipwreck: Insights into Aggregate Transport

The Ses Llumetes shipwreck, first century CE, at 2 m depth in the Balearic Islands near Porto Cristo, on the coast of Mallorca, Spain, preserves a loosely consolidated, olive-gray deposit, up to 25 cm thick, dominated by pumiceous tephra, commonly armored with fine ash (**Figures 3** and **7*a*,*b***). Bulk compositions of the shipwrecked pumice fall within the compositional field of Baia tephra deposits, although glass analyses show more dispersion (**Figure 3*c***). REE of a single lapillus are nearly identical to pumice from Astroni crater (Tonarini et al. 2009), yet powders blended from smaller clasts more closely follow REE of Fondi di Baia tephra (Fedele et al. 2011) (**Figure 7*c***; **Supplemental Table 2**). The analyses suggest that pumiceous tephra from various Campi Flegrei deposits could have been combined for transport.

Curiously, pyroclastic deposits with phonolitic-tephriphonolitic compositions occur nearby, in the Columbretes archipelago (Aparicio & Garcia 1995, Anochea & Huertas 2020) (**Figure 3*c***; **Supplemental Figure 4**); REE diverge from Baia and Ses Llumetes tephra compositions, however (**Supplemental Figure 4*j***). The islands are mentioned by Strabo (1917) as *Ophiusa* [Ὀφιοῦσσα, i.e., serpents (3.5.1)] and by Pliny the Elder (1962, 3.14, 15.21, 35.59), noting an abundance of snakes—but with no mention of volcanic deposits. Roman builders seem to have disregarded the Columbretes deposits as sources of concrete aggregates. They also disregarded voluminous rhyolitic pumiceous tephra deposits of the Late Bronze Age eruption of Santorini volcano

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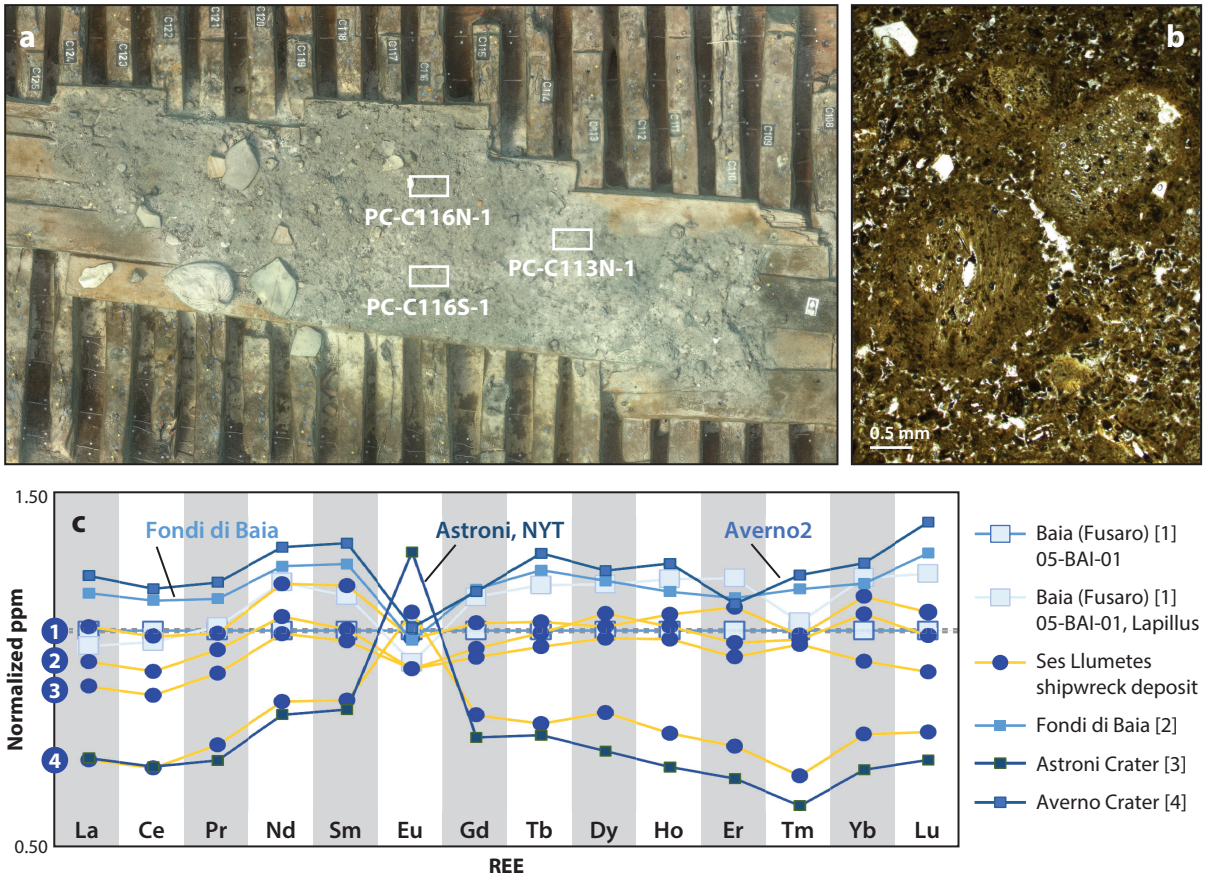


Figure 7

The Ses Llumetes shipwrecked pumiceous tephra deposit at Porto Cristo, Mallorca, first century CE (**Figures 3**; **Supplemental Figures 4–6**, **Supplemental Table 2**, **Supplemental Geochemical Data**). (*a*) Composite image of the timbered frame of the hull, tephra deposit, and sample locations; 19-PC-01 lies left of the image (Aragon et al. 2023). (*b*) Pumice clasts with accretionary fine-ash rims, plane polarized light. (*c*) REE studies of Campi Flegrei and Ses Llumetes pumice clasts normalized to Baia (Fusaro) pumice composition, Samples: ① 19-PC-01, ② PC-C113N-1, ③ PC-C116S-1, and ④ PC-116N-1. References: [1] Jackson et al. (2013a), [2] Fedele et al. (2011), [3] Tonarini et al. (2009), and [4] Di Vito et al. (2011). Abbreviations: REE, rare earth elements; NYT, Neapolitan Yellow Tuff.

Supplemental Material >

(Druitt et al. 2019); the rhyolite has greater SiO_2, less $Na_2O + K_2O$, and a different primary crystal assemblage and REE ratios (Druitt 2014). No trace of rhyolitic pumice has been detected in the Roman concretes (**Figure 3*c***).

The Ses Llumetes deposit also preserves clasts of pumiceous mortar, some with plaster coating (**Supplemental Figure 4*f***), and shards of different types of ceramics stored as cargo (**Supplemental Figure 5**). Marks on timbers suggest a connection with Ostia, near Rome (Aragon et al. 2023). The cargo seems to record the transport of Italic and Baetican goods, from Hispania, through the Balearic Islands as part of an intricate Mediterranean redistribution system (**Supplemental Figure 6**). Unit weights vary from 0.95 g/cm^3 for dry pumice to 1.53 g/cm^3 for seawater-soaked construction debris, so the deposit did not contribute heavy ballast. Perhaps the deposit cushioned the amphorae under transport (Brandon et al. 2014, pp. 223–33). Perhaps the *pulvis* and construction debris would eventually have been employed as concrete aggregate

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(**Supplemental Figure 4*g*,*h***). Mortars in Pompei post-dating the 62–63 CE earthquake contain reused mortar clasts, perhaps recycling the ruins of concrete structures (Miriello et al. 2018, Franceschini et al. 2024).

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5. RESILIENCE (*SOLIDITAS*) AT MILLENIAL SCALE

Recent research verifies some of the empirical observations and hypotheses for mechanisms that underlie resilience of *caementiciae structurae* recorded by Roman natural scientists (**Figure 8**) and reveals the chemical, mineralogical, and mechanical mechanisms underlying the longevity of the ancient concrete (**Figure 9**).

5.1. Roman Hypotheses: Energetic Inputs and Hydration Dynamics

Producing the reactive materials of marine and architectural concrete structures requires intensive energetic input (Vitruvius 1931, 2.5.1–2.5.3, 2.6.1–2.6.3) (**Figure 8**, labeled 1). Fire that generates calcination of limestone in kilns, which produces quicklime (*calyx*), and the Fire element that generates hydrothermal activity and explosive pyroclastic eruptions, which produce pumiceous tephra (*pulvis*), tuff (*tofus*), and sponge-stone pumice (*spongia*) aggregates, are hypothesized to create materials depleted in the Liquid element (Vitruvius 1931, 2.6.2–2.6.3). (This hypothesis refers to Four Elements theory, described in Vitruvius 1931, 2.2.1.) Through hypothesized acceptance of the Liquid element, the mixture hydrated with (sea)water is observed to quickly cohere and solidify (*solidantur*) (**Figure 8**, labeled 2). It grows hot with hypothesized latent heat (*latenti calore*), produced by a hot lack of moisture (*calida umoris*) suddenly being filled with water (*aqua*) (Vitruvius 1931, 2.6.4). Perhaps Roman thought considered a potential lying inherent, or *latent*, within a substance that could be realized or released with application of certain empirical protocols, such as adding seawater to lime and *pulvis*. In the sense of this *latent* potential for reactivity, with consequent change in free energy of the system, the physical manifestation would be a reaction product and evolution of heat (L.W. Hobbs, personal communication 2025). A calculation of the temperature rise in the *Baianus Sinus* pila associated with exothermic, or *latent*, heat released mainly through formation of C-A-S-H binding phase finds that elevated temperatures, 85–95°C, persist in the center of the pila for 1–2 years (Jackson et al. 2013a).

Extended hydration (**Figure 8**, labeled 3) is implied for marine concrete (Vitruvius 1931, 2.6.1) and hypothesized for architectural concretes that "maintain themselves in their [own] moisture (*umore*) into very old age (*ad summam vetustatem*)" (Vitruvius 1931, 2.8.6). This fluid, also described as vital sap or juice (*sucum*), must remain in the mortar (*materia*) to maintain cohesion in the concrete wall (Vitruvius 1931, 2.8.2; Brandon et al. 2014, p. 20). Pliny the Elder (1962), 100 years later, described the marine concrete as "a single stone mass (*lapidem unum*) impregnable to the waves and every day stronger" (35.166) (**Figure 8**, labeled 4), validating the implication of extended hydration hypothesized by Vitruvius and possibly foreshadowing the diagenetic processes that produce zeolite cements in volcaniclastic rocks (Passaglia et al. 1990). The natural consolidation of *puteolanus pulvis* in the presence of water (*si aquam attigit*) to become rock (*saxum*) is also described by Seneca (1971, 3.20.3–3.20.4). Young oceanic basaltic tephra at Surtsey volcano, Iceland, lithifies to form tuff with Al-tobermorite and phillipsite mineral cements at 30–140°C (Jakobsson & Moore 1986, Montesano et al. 2023, Jackson et al. 2024b).

5.2. Ongoing Cycling of Cementitious Products

Modern binding systems that activate aluminosilicate components, such as natural pozzolans and supplemental cementitious materials, are designed to follow distinct reaction pathways (Provis & Bernal 2014). High calcium (and/or magnesium) alkali earth metal systems of blended Portland

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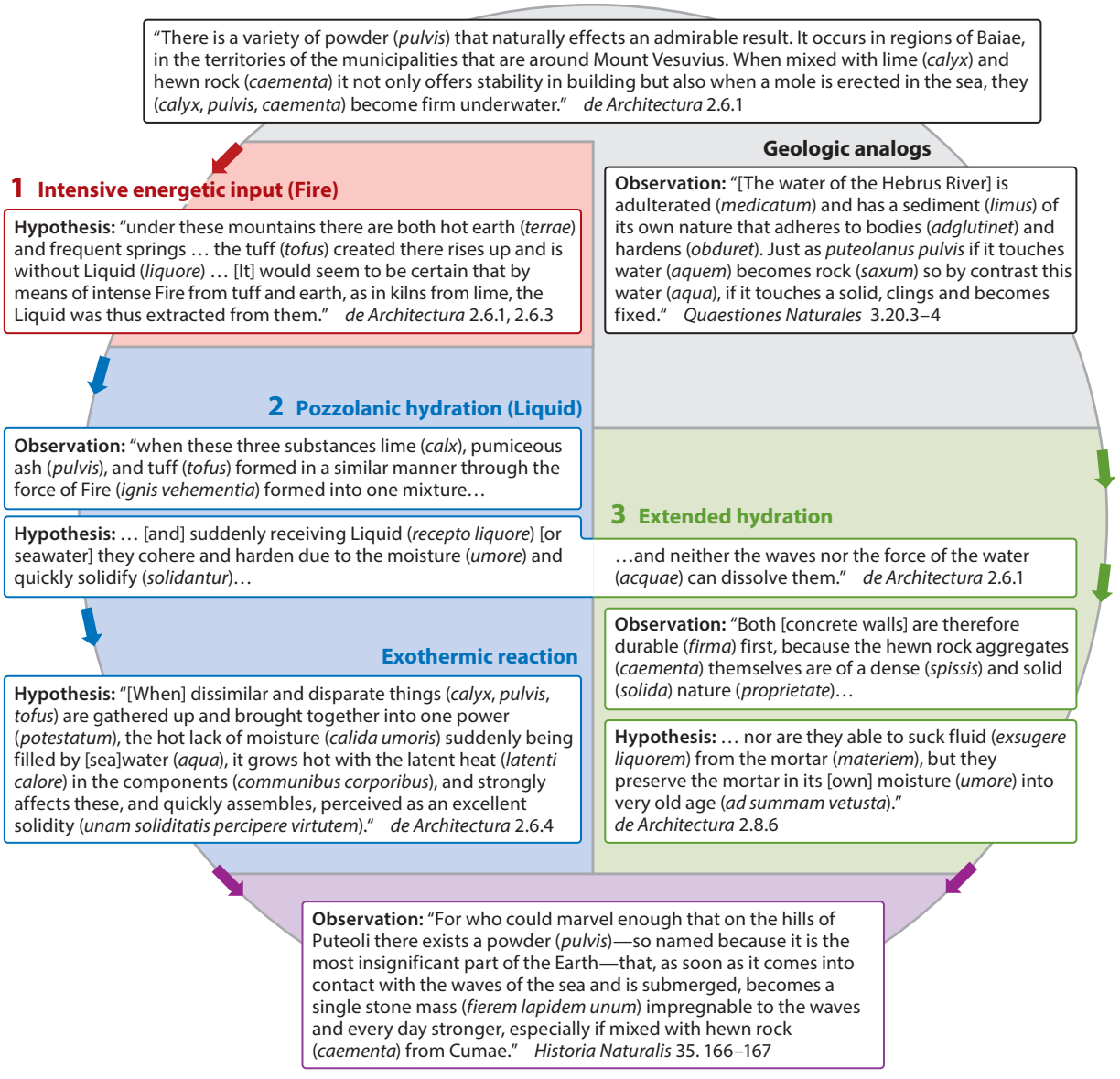


Figure 8

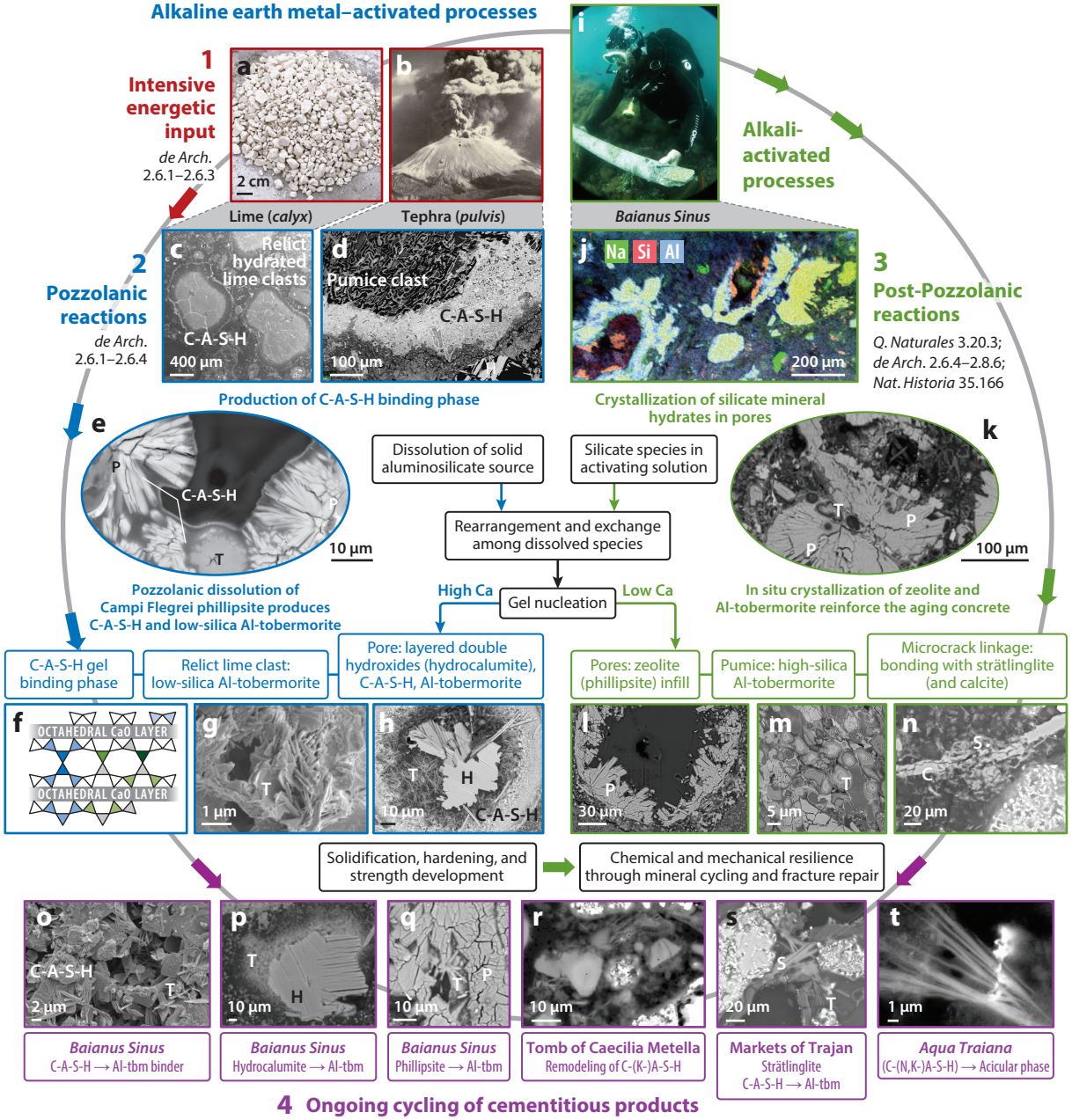
Descriptions of materials, cementitious processes, and hypotheses for mechanisms underlying the resilience of concrete structures, *caementiciae structurae*, by Roman natural scientists: Seneca (ca. 4 BCE–65 CE), Vitruvius (ca. 90–20 BCE), Pliny the Elder (23/24–79 CE). Geochemical, mineralogical, and material studies (**Figures 3–7** and **9**) validate these ancient records of "scientific knowledge (*scientia*) acquired [through] deductive reasoning (*ratiocinatione*) and skillful craft (*fabrica*)" (Vitruvius 1931, 1.1), which provide guiding principles for building with *caementiciae structurae* (Jackson & Kosso 2013, Lefas 2023).

cement concretes produce C-(A)-S-H–type gel and layered calcium aluminate phases. Conversely, low calcium systems of alkali-activated or geopolymer concretes (Kriven et al. 2024) produce an alkali aluminosilicate–type gel [sodium-aluminate-silicate-(hydrate)—N-A-S-(H)] and crystallization of zeolites. The stability of the reaction products constituting the solid binder is thought to

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be a key factor governing the long-term performance of the cement or concrete (Myers et al. 2014).

Remarkably, the binding system of *caementiciae structurae* that was envisioned by Roman natural scientists and practiced by Roman builders activates both reaction pathways shown by the Provis & Bernal (2014) diagram (**Figures 8** and **9**). Calcination of limestone and pyroclastic eruptions



(*Caption for* ***Figure 9*** *appears on following page*)

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Figure 9 (*Figure appears on preceding page*)

Diagram of reaction pathways illustrating ongoing cementitious processes in *caementiciae structurae*, as described in extant Roman texts (**Figure 8**) and placed within the context of modern concepts of pozzolanic and alkali-activated processes (based on the diagram of Provis & Bernal 2014). (*a*,*b*) **1** Intensive energetic inputs, through calcination of limestone (900°C) and explosive pyroclastic eruptions (800–1,000°C), produce reactive materials (Vitruvius 1931, 2.6.1–2.6.3). (*c*–*b*) **2** Pozzolanic reactions produce cementitious binding hydrates and early cohesion of concrete (Vitruvius 1931, 2.6.1–2.6.4). (*i*–*n*) **3** Chemical interactions of the concrete with fluids in the massive structures (Vitruvius 1931, 2.8.6) produce post-pozzolanic mineral cements, with analogies to diagenetic processes in rock (Seneca 1971, 3.20.3; Pliny the Elder 1962, 35.166–35.167). (*o*–*t*) **4** The cementitious products themselves react to produce reinforcing silicate mineral cements. Credits: *a*,*n*,*p*, M.D. Jackson; *b*, Vesuvius eruption 1944, photo from the Smithsonian Institution (public domain); *c*,*d*,*h*, Brandon et al. (2014); *e*,*j*,*k*,*l*,*m*,*q*, Jackson et al. (2017); *f*,*g*, Jackson et al. (2013a); *i*, D Bartoli, ROMACONS project, *o*, Elsen et al. (2022); *r*, Seymour et al. (2021), *s*, reproduced from Jackson et al. (2024a) (CC BY 4.0); *t*, reproduced from Medeghini et al. (2024) (CC BY 4.0). Abbreviations: C-A-S-H, calcium-aluminum-silicate-hydrate binding phase; C, calcite; H, hydrocalumite; P, phillipsite; S, strätlinglite; T, Al-tobermorite.

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(**Figures 8** and **9*a*,*b***, labeled 1) generate the raw materials of pozzolanic reaction (**Figures 8** and **9*c*–*e***, labeled 2). In marine concrete specifically, activation with calcium hydroxide produces dissolution of aluminosilicate phases in tephra aggregate. Rearrangement and exchange among the dissolved species produce C-A-S-H binding phase in relict clasts of hydrated lime, the cementing matrix (**Figure 9*c***), and the accretionary ash and vesicles of pumice clasts, which may contain Campi Flegrei phillipsite (**Figure 9*d*,*e***). Low-silica Al-tobermorite crystallizes in relict clasts of hydrated lime (**Figures 4*g*,*h*** and **9*c*,*g***), and hydrocalumite crystallizes in pores (**Figure 9*d*,*h*,*p***). Al-tobermorite has a cross-linked structure (**Figure 9*f***) that incorporates alkali cations, similar to modern C-(N,K-)A-S-H syntheses (Jackson et al. 2013a,b; Myers et al. 2015; L'Hôpital et al. 2016). These phases produce solidification, hardening, and strength development, perhaps at briefly elevated temperature (Jackson et al. 2013a). A transition to alkali-activated processes occurs after calcium hydroxide is largely consumed (**Figures 8** and **9**, labeled 3). Low-temperature interactions with seawater-derived fluids produce silicate species in activating solutions within the heterogeneous pore and permeability structure of the concrete (**Figure 6**). Phillipsite crystals precipitate from alkaline fluids (**Figures 8** and **9*j*–*l***); Al-tobermorite crystals with elevated Al, Na, and K precipitate in pumice vesicles (**Figures 4*i*** and **9*m***; **Supplemental Figure 3**). Silicate minerals, such as strätlingite, also precipitate within microcracks and microcrack linkages (**Figure 9*n***).

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The cementing phases may themselves react to further remodel the concrete fabric (**Figures 8** and **9*o*–*t***, labeled 4). In marine concrete, Al-tobermorite replaces C-A-S-H in the cementing matrix (**Figures 4*g*,*h*** and **9*o***), hydrocalumite dissolves and Al-tobermorite crystallizes in pores (**Figure 9*h*,*p***), and authigenic phillipsite dissolves and Al-tobermorite crystallizes in micropores (**Figure 9*q***). In architectural concrete, C-A-S-H remodels itself to form tendrils and elongate plates (**Figures 4*c*,*d*** and **9*r***); Al-tobermorite crystallizes with strätlingite in C-A-S-H domains (**Figure 9*s***); and further acicular phases form and evolve, as in the *Aqua Traiana*, 109 CE, which continues to carry water from Lago Bracciano to Trastevere in Rome (**Figure 9*t***; **Supplemental Figure 1*a***).

5.3. Processes of Transformational Toughening

Transformation toughness refers to an increase in the fracture toughness of a material at the tip of an advancing crack, which is the direct result of a stress-induced phase transformation that absorbs energy and resists crack propagation. Zirconia, for example, can undergo a phase transformation from tetragonal to monoclinic structure under stress, effectively absorbing the energy that would otherwise cause the crack to further propagate (Hannink et al. 2010). The resilience of *caementiciae structurae* may derive from an analogous process, which does not involve a direct phase transformation but instead involves the transformational effect of a cementitious fabric

that evolves in complexity over time (W.M. Kriven, personal communication 2025). Intricate assemblages of cementing hydrates in microstructures that derive, in part, from the heterogeneous permeability structure of the concrete (**Figures 4**, **5*c***, **6**, and **9**) may mechanically interfere with crack propagation. These obstacles may cause deflection at the tip of an advancing crack, increasing fracture energy required to create new crack surface area, and further toughening the concrete through extending its ability to absorb energy and plastically deform before fracturing (van Mier 1997, Bažant et al. 2009). Arc-bending test experiments of a Markets of Trajan mortar replica provide an initial validation of this hypothesis through measured and computed increases in the G_F required to propagate microcracks at 90 and 180 days of hydration, when strätlingite crystals toughen interfacial zones and the cementing matrix (**Figure 5*a–c***).

6. ANCIENT PRINCIPLES FOR MODERN SELF-SUSTAINING CONCRETE

6.1. Responses of Concrete Structures to External Influences

The *caementiciae structurae* of extant monuments in Rome have resisted structural-scale failure during at least six episodes of moment magnitude 6.7–7 ground shaking events associated with slip on Appennine fault systems 80–130 km to the northeast (Frepoli et al. 2010, Galli & Molin 2014), as well as potential chemical decay associated with repeated inundations with Tiber River floodwaters (Aldrete 2007). The concrete structures contain common macroscale fractures, with rough surfaces that link by complex segment overlap and bridging (Jackson et al. 2014), yet the monuments maintain structural functionality as residences, museums, offices, churches, and aqueducts. Sectional failure of the Colosseum, 70–90 CE, Baths of Caracalla, ca. 215 CE, and Basilica of Maxentius, ca. 313 CE, for example, mainly occurred in Late Antiquity or the Middle Ages through subsurface instabilities; problematic structural design; removal of marble and travertine dimension stone, columns, and cladding; and absence of regular maintenance (Funiciello et al. 1995, Samuelli Ferretti 2005, Galli & Molin 2014).

The capacity of the architectural concretes to fracture at macroscale but remain intact at structural scale results from long-term interactions between architectural form and material responses to static loads and seismic ground shaking (e.g., Brune & Perucchio 2012). For example, a series of concrete lateral arches connect the 8.6-m span of the central vault of the Great Hall of the Markets of Trajan to the surrounding superstructure (**Figure 1*b***); the elevated location of the arches was perhaps intended to mitigate both static and seismic loadings. Selections of silt- to gravel-sized Pozzolane Rosse scoriaceous tephra (**Figure 2*a–c***; **Supplemental Figure 1*a***) and cobble-sized tuff (and ceramic) *caementa* as reactive aggregates are important components of the chemical and mechanical toughening mechanisms that reinforce these structures (**Figures 1*i*** and **4**). An initial pozzolanic system evolved to post-pozzolanic processes, which created further reinforcements as strätlingite and Al-tobermorite mineral cements (**Figures 4*e*,*f*** and **9*s***) and remodeled poorly crystalline phases (**Figures 4*c*,*d*** and **9*r***) (Jackson et al. 2009, 2014, 2024a; Seymour et al. 2021). Slow permeation of Tiber River floodwater into walls and foundations may have benefited these processes (**Figure 6*a*,*c***).

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The *caementiciae structurae* of extant marine harbors in the Bay of Pozzuoli have absorbed energy from $M_d < 4$ seismic ground shaking associated with activity at Campi Flegrei caldera, as well as seafloor displacements produced by secular subsidence superimposed on periods of ground uplift (Costa et al. 2022, Orsi 2022); current uplift and subsidence phases have an axisymmetric shape centered in Pozzuoli harbor (Pino et al. 2025). Pliny the Elder (1962) described inundation of the sea, as floods, and subsidence of the land during earthquake activity, as well as the sudden motion of waves in the sea (2.82, 2.83, 2.86; Hughes 2013). The *Caesarea Maritima* harbor (**Figure 1*f*,*g***),

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for example, was subjected to strong wave forces associated with distant earthquakes in first century CE and sixth century CE, as well as long-term subsidence of the seafloor (Reinhardt et al. 2006, Galili et al. 2021).

The force of impact of strong waves on concrete structures, generated by typhoons, hurricanes, winter storms, and earthquakes over the centuries, could be close to tens of thousands of metric tons per square meter in large events (Mehta 1991, p. 25). Strong wave action affects structures mainly in the supratidal zone, where the concrete is also vulnerable to abrasion by sand and gravel suspended in seawater. Maritime concretes are now systematically reinforced with steel but may suffer pitting corrosion of steel by chlorides, salt weathering, disaggregation, and spalling (Mehta 1991, pp. 64–9; Hren et al. 2024). The reinforced structure must therefore remain highly impermeable to seawater inflow. High-performance blended cement concrete with supplemental cementitious materials has very low permeability, 10^{-17}–10^{-18} m^2 (**Figure 6*e***), yet steel-reinforced structures are vulnerable to corrosion through fracture and inflow of seawater (Mohammed et al. 2003).

By contrast, ancient Roman designs for concrete structures in diverse hydrological environments produce silicate cementitious phases through beneficial interactions of fluids with reactive aggregates, which reinforce cohesion over a long history of service life. In *Caesarea Maritima* marine concrete, poorly crystalline magnesian aluminosilicate binding phases, M-(A)-S-H, form from pore solutions derived from combustion residues of organic materials, dolomitic rock, and shell fragments (Secco et al. 2019, 2022); there are also disordered low-calcium C-A-S-H phases. In a first-century CE well-cistern in the lagoon of Venice, early calcium-bearing binding phases are also replaced by M-(A)-S-H phases (Dilaria et al. 2024). In the *Piscina Mirabilis*, the cistern constructed near Miseno, Italy, as the terminal reservoir of the *Aqua Augusta*, Al-tobermorite crystals in micropores derive from long curing of concrete containing trachytic tuff and ceramic reactive aggregates (Rispoli et al. 2020) (**Figure 1*d***; **Supplemental Figures 1*b*** and **3**). Mortars of the *Aqua Virgo* aqueduct, ca. 19 BCE, near Rome contain blade-like cementitious gels (Calzolari et al. 2023). The mortar of the *Aqua Traiana* aqueduct, 109 CE (**Figure 9*t***), records C-(N,K-)A-S-H gel with an acicular phase as well as authigenic zeolite in scoria vesicles (Medeghini et al. 2024). And 11 Å tobermorite and 14 Å piomberite occur in the mortar of *Ponte Augusto* at Narni, 27 BCE (Cantisani et al. 2002). Slow infiltration of fluids into the reactive fabric of these concretes evidently renews cohesion and, perhaps, restores structural functionality to a more resilient state than that of the early-age concrete.

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The average measured uniaxial compressive strength, 6 MPa, and Young's modulus, 7,320 GPa, of 9-cm diameter drill core specimens of the marine concrete are similar to that of Imperial era conglomeratic concrete in the monuments of Rome, 6 MPa and 2,560 MPa, respectively, measured in 1 m^3 reproductions with ceramic *caementa* (Samuelli Ferretti 1995; Brandon et al. 2014, pp. 174–80). Typical uniaxial compressive strength of ordinary Portland cement concrete is about 30 MPa, and Young's modulus of hardened paste is 10–30 GPa.

Roman designs did not require high compressive strength for structural functionality in broad reefs, breakwaters, moles, and wharves in the sea—or in massive walls, piers, and substructures on land. The principal performance requirements of the ancient marine structures were to resist the impact force of strong waves and remain anchored on the seafloor, to maintain resistance to abrasion in the supratidal zone, to preserve cohesion during moderate magnitude earthquake ground shaking, and to gain in resilience over time so that "neither the force of the waves nor the effect of water can dissolve them" (Vitruvius 1931, 2.6.1). Laboratory measurements of low compressive strength do not adequately represent the extraordinary mechanical resilience of the ancient structures.

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6.2. Marine and Coastal Concrete Infrastructure

Roman builders integrated material resilience, beneficial hydrologic interactions with seawater, ecological compatibility, and geomorphological continuity to maintain the structural functionality of *caementiciae structurae* in marine environments for two millennia. The concrete mix was formulated with seawater to deliberately enhance chemical compatibility with the marine environment, as well as to conserve freshwater resources. Infiltration of seawater through the complex permeability structure of the concrete (**Figure 6*b*,*d***) produces chemical interactions with reactive pumiceous aggregate (**Figure 2*d–f***) that renew cohesion through long-term growth of silicate mineral cements (**Figures 4*g–i*** and **9*j–q***). The external surfaces of submerged structures have 10–20-cm-thick marine biological and mineralogical encrustations, including algae, seaweed, sponges, worm casts, and carbonate minerals, which create a protective surface and symbiotically enhance infrastructural resilience (Brandon et al. 2014, pp. 8, 40, 275, 277). The structures were commonly built as reefs that follow the natural topography of the seafloor and parallel coastal structures derived from longshore sediment transport, thus maintaining geomorphological continuity over the centuries.

A framework of conglomeratic rock and ceramic aggregate(s) reinforces the massive structures. Wood formwork may continue a short distance below the seabed, yet the relatively fragile planks were likely not intended to anchor the large and heavy masses (Brandon et al. 2014, pp. 189–222). The ductile response measured in Roman analog marine concrete arc-bending tests (**Figure 5*d***) suggests that the structures may resist fracture more effectively than Portland cement concrete, which has substantially greater early compressive strength yet typically degrades monotonically until maintenance and/or replacement, with greater rates of decrease caused by cracking, seawater ingress, and corrosion of steel reinforcement (e.g., Torres-Acosta et al. 2007). Cohesion of extant Roman marine structures remains robust, yet eventual disaggregation would return the conglomeratic reinforcing framework to the seafloor as benign rock sediment.

7. FUTURE STUDIES AND APPLICATIONS

An inferred modern production process for Roman concrete would emit as much greenhouse gas per volume as conventional cement-based concrete (Martinez et al. 2025). This high initial energy demand, recognized by Romans (**Figure 8**), and its negative environmental impacts could be mitigated by increased durability and longevity, requiring less maintenance and replacement over time. Nevertheless, direct adoption of *caementiciae structurae* in conventional concrete designs would be impeded by slow early strength gain, pore fluid pH < 12.5 insufficient to protect steel reinforcement from corrosion, and low compressive strength that requires large structural elements (Martinez et al. 2025).

Concrete structures are critical to the repair of damaged shorelines, aging waterways, and mitigation of rapid hydrological change associated with sea level rise and climate instabilities, storm surge, and coastal erosion (US Army Corps of Engineers 2021). Well-integrated saltwater concrete structures would ideally produce long-lasting coastline reinforcements that create favorable environments for plant, invertebrate, and fish communities (Suedel et al. 2022). This is in contrast to hard, armored, conventional Portland cement concrete structures that disrupt coastal ecosystems and require costly maintenance and/or replacement, due in part to corrosion of embedded steel reinforcement; innovations in seawater-mixed Portland cement concretes currently show unpredictable durability outcomes (Aghajanzadeh et al. 2026).

Elements of Roman material design for *caementiciae structurae* could provide a paradigm to be incorporated in modern materials for shoreline concretes. Reef-like structural designs

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(Brandon et al. 2014, pp. 122–40; De Graauw 2022) would balance material tensile capacity with structural dimensions to achieve an effective safety factor; large coarse aggregate, or *caementa*, would increase toughness and provide a self-reinforcing framework without steel or glass fiber rebar; and dynamic chemical interactions among reactive aggregates and pore fluids would beneficially remodel and toughen cementitious fabrics. Fracture could induce a beneficial change to material permeability (e.g., Heap & Kennedy 2016) as fluid ingress interacts with reactive aggregates to generate cementitious products that repair and restore the material continuum. Absence of steel– or glass fiber reinforced polymer–reinforcement would reduce initial material fabrication costs and emissions, as well as construction costs, corrosion, and ecological end-of-life costs (Younis et al. 2018, Morales et al. 2021). Viable sources of reactive volcanic rock and synthetic glass foam aggregates for large-scale structures currently remain elusive, however.

Insights gained through further archaeometric investigations of ancient mortars and concrete would clarify interactions among the material characteristics of reactive aggregate materials, environmental conditions, and time that influence dynamic evolution of post-pozzolanic cementitious hydrates and microstructures, as well as the interactive material and mechanical behavior of specific Roman structures. These studies could further inform the resilience of novel lime-clay binding systems, for example, in which a pumice addition mitigates damaging alkali silica reaction through crystallization of calcium-bearing minerals in vesicles, which then evolve to phillipsite and chabazite zeolite (Tahsin et al. 2025), similar to the post-pozzolanic processes of *caementiciae structurae*. Reactions of synthetic calcium (alumino)silicate glasses, basaltic glasses, borosilicate glasses, and recycled domestic glass with aqueous solutions at varying pH (e.g., Snellings 2015, Jantzen et al. 2017, Malferrari et al. 2022) have produced various hydrated, crystalline precipitates, including Al-tobermorite. The cross-linked structure and Al^{3+} substitutions for Si^{4+}, balanced by Ca^{2+}, Na^{+}, and K^{+} cations in the interlayer of Roman syntheses of Al-tobermorite crystals (Jackson et al. 2013a,b, 2017), could provide clues to creating low-temperature crystallization pathways for cation change (e.g., Komarneni et al. 1987) in alkali-activated concretes for cementitious barriers and waste encapsulations.

8. SUMMARY

Advanced analytical methods and materials testing validate Roman hypotheses for mechanisms of self-sustaining resilience in the *caementiciae structurae* of ancient architectural and marine construction. An early pozzolanic system activated with hydrated lime consolidates and strengthens the concrete, mainly through production of C-A-S-H binding phase in a complex cementing matrix. Low-temperature interactions of reactive volcanic rock aggregates with pore fluids then generate a post-pozzolanic, alkali-activated system that toughens the concrete through production of silicate mineral cements and, in some structures, remodeling of the early binding fabric. A heterogeneous permeability structure, derived in part from cobble-sized *caementa* and a broad size distribution of reactive mortar aggregates, principally potassic scoriaceous tephra and trachytic pumiceous tephra, facilitates these processes. A survey of *pulvis* pumiceous tephra employed in marine and coastal concrete structures, first century BCE to third century CE, and the Ses Llumetes shipwreck reveal a broad diaspora of tephra aggregate transported from the Campi Flegrei volcanic district to structures throughout the Mediterranean region. Incorporation of elements of Roman material design for *caementiciae structurae* in modern materials with reactive rock aggregates for shoreline and coastal concrete structures could improve structural functionality through beneficial hydrologic interactions with saltwater, self-sustaining material resilience, ecological compatibility, and continuity with coastline geomorphology.

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DISCLOSURE STATEMENT

M.D. Jackson is a co-inventor of US patents 12297147, 12325662, 12421168, which relate to self-sustaining, reactive cementitious systems.

ACKNOWLEDGMENTS

The research presented in this review reflects the invaluable contributions of our numerous colleagues and collaborators who have advanced modern knowledge of ancient Roman concrete over the past 20 years. We extend special thanks to John Oleson, Linn Hobbs, Trudy Kriven, and Gilberto Artioli, who provided new insights into the nature of *caementiciae structurae* for this review. Barbara Nash provided analyses of volcanic glass at the Electron Microprobe Laboratory in the Department of Geology and Geophysics at the University of Utah. Undergraduate students Damien Benot and Colin Sanchez at the École et Observatoire des Sciences de la Terre (France) created the permeability maps (under the supervision of M.J. Heap), with assistance from undergraduate student Isabelle Bowers, University of Utah. Philip Brune, Charles Weiss, Jedediah Burroughs, Juan Carlos de Obesa, and Hans-Rudolf Wenk provided research support; Concetta Rispoli, Cristina Belfiore, Heng Chen, and Laura Medeghini provided data and images from ancient concretes (**Figures 2*d–f*** and **4*a*,*b*,*g*,*h*** and **9*t***). The Sovrintendenza Capitolina ai Beni Culturali di Roma Capitale, Soprintendenza Speciale di Roma, Archeologia e Belle Arti e Paesaggio, Soprintendenza Archeologia di Roma, Parco Archeologico dell'Appia Antica, and ROMACONS drilling project provided generous access to samples of ancient concretes. The Boulouki project provided access to the Agrilia settlement on Therasia. The Montana State University Imaging and Chemical Analysis Laboratory provided microscopy imaging and analysis. M.D. Jackson acknowledges funding from US Department of Energy (DOE) ARPA-E award DE-AR0001139, United States Army Corps of Engineers Engineering with Nature® grant ERT-23-14, the Loeb Classical Library Foundation at Harvard University, and the Department of Geology and Geophysics at the University of Utah. This research used beamline 12.3.2, which is a resource of the Advanced Light Source, a US DOE Office of Science User Facility under contract DE-AC02-05CH11231. M.D. Jackson is a principal in Caementa, LLC, which has provided materials to aid this research. M.J. Heap acknowledges support from the Institut Universitaire de France (IUF) and European Research Council Synergy Grant ROTTnROCK (ERC-ROTTnROCK-101118491). A.G. Sinner acknowledges funding from the Social Sciences and Humanities Research Council of Canada, award 430-2020-00317, *The 'Ses Llumetes' Wreck (Porto Cristo, Majorca): Trade, Shipbuilding and Technology in the Western Mediterranean during the first century CE* project, and the University of Victoria for an Internal Research/Creative Project Grant. We thank Sebastia Munar, Javier Rodríguez, and the Instituto Balear de Estudios en Arqueología Marítima for conducting the underwater excavations and sampling.

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